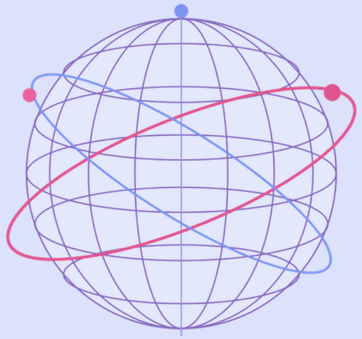
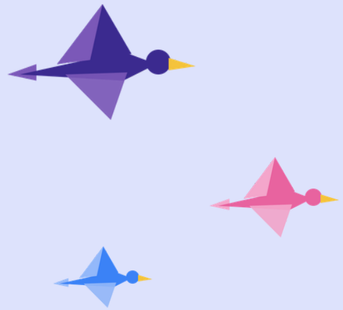
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
Noise-Aware Adaptive Quantum Error Correction

TRACK
OFFLINE

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Problem Statement

Sensitivity to Noise: *Near-term quantum systems are highly vulnerable to bit-flip, phase-flip, and readout errors, which degrade computational accuracy.*

Inflexible Correction: *Standard error-correction methods operate statically, failing to optimize performance across dynamic or mixed noise profiles.*

Heavy Resource Overhead: *Error correction requires extra physical qubits and deeper circuits, which can introduce more noise than they eliminate.*

The Core Challenge: *Developers need an automated, data-driven framework to answer:*
"Which error-correction strategy maximizes circuit reliability within a given qubit budget and noise regime?"

Objectives & Novelty

Objectives

Automated Noise Analysis: Evaluate ideal, noisy, and corrected quantum execution across configurable Pauli X/Z, readout, and gate error parameters.

Rigorous Benchmarking: Compare uncorrected baselines against 3-qubit bit-flip and phase-flip repetition codes across both \vert{}1\rangle and \vert{}+\rangle test states.

Novelties

Adaptive Decision Engine: Recommends optimal strategies using a dual-run evaluation (calibration vs. validation) to eliminate overfitting.

Resource Cost Penalties: Applies an explicit physical-qubit budget and user-configured quality-cost trade-offs.

Honest Fallback Mechanism: Automatically recommends the uncorrected baseline whenever correction overhead degrades total circuit performance.

Q-Adapt does not assume "error correction always wins" — it scientifically identifies the exact noise threshold and resource boundary where error correction is genuinely justified.

Solution & Quantum Advantage

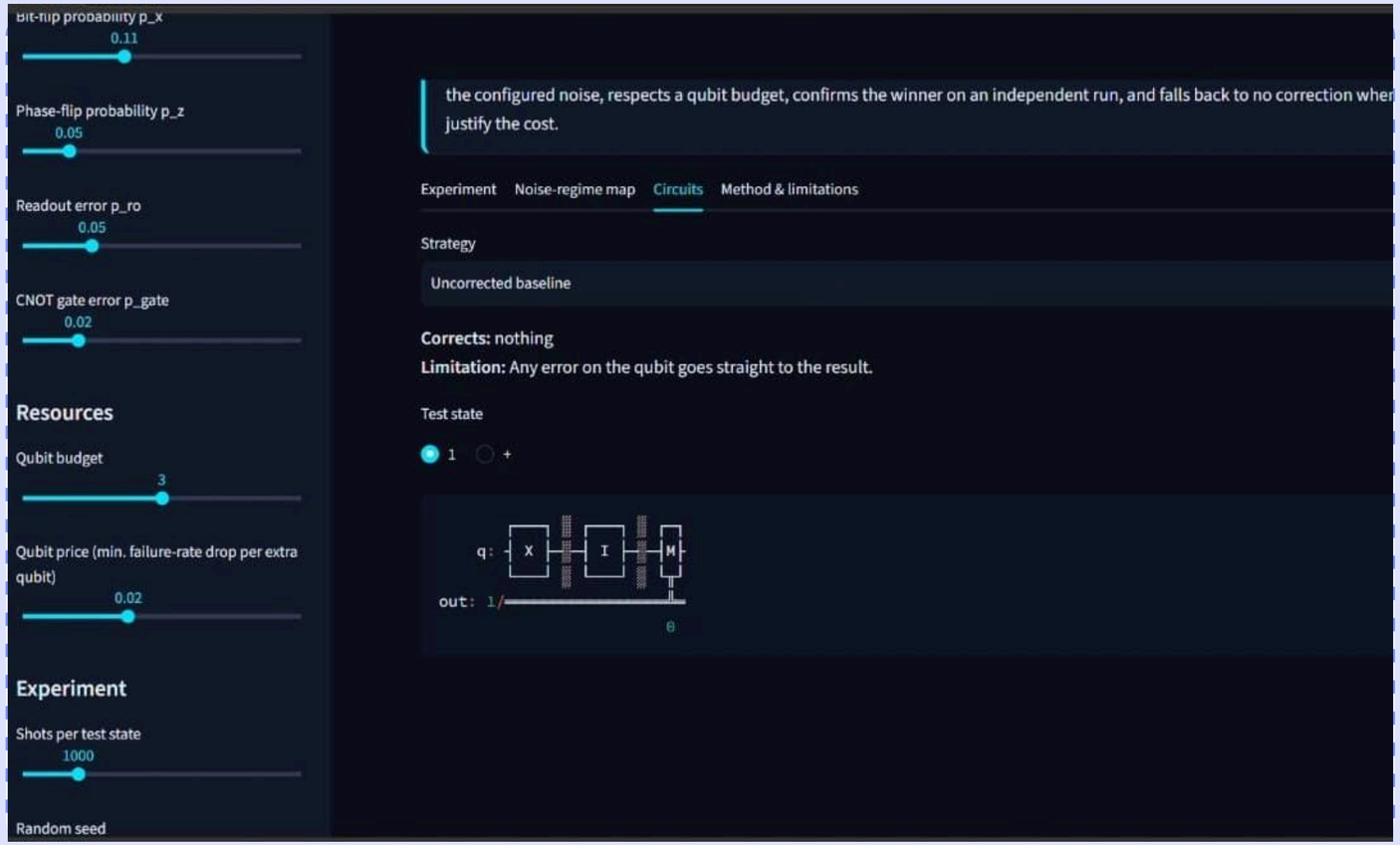
Solution Workflow

Configure Noise: Set custom probabilities for bit-flip (p_x), phase-flip (p_z), readout (p_{ro}), and CNOT gate (p_{gate}) errors.Generate Test Circuit: Synthesize target logical test states (\vert{}1\rangle and \vert{}+\rangle) across baseline and encoded schemes.Simulate Strategies: Run uncorrected baseline, 3-qubit bit-flip code, and 3-qubit phase-flip code in parallel using Qiskit Aer.Calculate Reliability & Resource Metrics: Compute logical failure rates with 95% confidence intervals, physical qubit counts, circuit depth, and total gate overhead.Recommend Strategy: Filter strategies by user-specified qubit budget and select the top performer verified on independent validation data.

Quantum Advantage

Simulated Insights: Demonstrates how Qiskit's circuit-level noise modeling helps analyze quantum error propagation and logical state protection without wasting physical QPU time.Resource Optimization: Prevents quantum developers from over-allocating physical qubits when noise levels do not justify correction overhead.

Algorithm & Circuit Design



***Logical Failure Rate**:*
Logical Failure Rate = Incorrect Decoded Shots / Total Shots
***Success Probability**:*
Success Probability = Correct Decoded Shots / Total Shots = 1 - Failure Rate
Resource Overhead Tracking:
***E**valuated using physical qubit count, circuit depth, and total gate count.*
***Dual-Run Engine**:*
Selection algorithm chooses the top strategy using calibration data and verifies statistical reliability on independent validation data with 95% Wilson score confidence intervals.

Implementation & Tech Stack

- ***Python & Qiskit****: We build the encoding and decoding circuits in Qiskit, transpile them into a common gate set, and correct errors with a majority-vote decoder.*
- ***Qiskit Aer****: We run every experiment on Aer's density-matrix simulator with noise models, and fixed random seeds make each run reproducible.*
- ***Custom Noise Models****: We simulate bit-flip, phase-flip, CNOT depolarizing and read-out errors, and the user can adjust the strength of each one.*
- ***Adaptive Decision Engine****: A transparent, rule-based engine removes strategies that exceed the qubit budget, keeps only those that significantly beat the baseline, and picks the most reliable option at the lowest qubit cost.*
- ***Streamlit Interface****: We built a dark-themed interactive dashboard that shows Plotly charts, circuit diagrams and a noise-regime map, and lets the user export results as CSV.*
- ***Testing & Validation****: We checked the results with 35 automated tests, including comparisons against theoretical error-rate formulas, and confirmed each recommendation on separate held-out runs.*

Results & Analysis



	label	failure_rate	ci_low	ci_high	success_prob	improvement_vs_baseline	fail |1>	fail |+>	qubits	depth	gates	eligible
0	Uncorrected baseline	0.1375	0.1231	0.1533	0.8625	0.0000	0.1780	0.0970	1	3	1	☑
1	3-qubit bit-flip code	0.1825	0.1662	0.2000	0.8175	-0.3273	0.1550	0.2100	5	14	12	☐
2	3-qubit phase-flip code	0.2130	0.1956	0.2315	0.7870	-0.5491	0.1300	0.2960	5	16	18	☐

Negative improvement = worse than baseline. Table shows the VALIDATION run; the engine chose using the CALIBRATION run and then confirmed here.

Reproducibility

```
{
  "p_x": 0.11,
  "p_z": 0.05,
  "p_ro": 0.05
```

Live Metrics:
Success Probability: Measured over 1,000–4,000 shots.
Logical Failure Rate: Evaluated with 95% confidence intervals.
Improvement: Measured as % drop in failure rate.
Resource Cost: *Tracks qubits (1 vs 5), depth, and gates.*
Dual Validation: Selection is made on calibration data and confirmed on independent validation runs.

Impact & Real-World Applications

- ***Pre-Execution Optimization:*** *Q-Adapt evaluates error-correction options in simulation before any job is sent to a costly physical QPU.*
- ***Qubit Budget Enforcement:*** *It respects strict qubit limits, so it never recommends a strategy that needs more qubits than are available.*
- ***Honest Recommendations:*** *It recommends no correction when none helps. This avoids wasting qubits and gates on correction that adds more error than it removes.*
- ***Matches the Code to the Noise:*** *It shows that a code must match the noise type. In our tests, the phase-flip code raised the failure rate under bit-flip noise from 7.4% to 16.3%.*
- ***Scales With the Noise:*** *In our results, the engine moved from the 3-qubit to the 5-qubit code as bit-flip noise grew, and it fell back to no correction when gate errors were too high.*
- ***Educational Insights:*** *It makes the trade-off between gate noise and code overhead easy to see through charts and a noise-regime map.*
- ***Reproducible and Transparent:*** *Every number comes from a real simulation with fixed seeds and confidence intervals, and every decision comes with a written explanation.*
- ***Future QPU Integration:*** *The framework can be extended to use noise measured on IBM Quantum backends through Qiskit Runtime. This is planned work, and today all noise is simulated.*
- ***Key Takeaway:*** *Q-Adapt makes error correction resource-aware, showing when correction helps, when it hurts, and when it costs too much.*

Challenges & Future Scope

Technical Challenges

- ***Idealized Noise:*** *Our noise models apply errors during a storage window. They do not capture hardware crosstalk, decoherence or leakage, and all noise is simulated.*
- ***Code Scope:*** *The 3- and 5-qubit repetition codes protect against one error type at a time (X or Z), not general noise. Under mixed noise, no single code helped.*
- ***Overhead Penalty:*** *Deep correction circuits need high gate fidelity to beat the baseline. In our tests, a CX error of just 0.6% made the 3-qubit decoder lose to doing nothing.*
- ***Costly Decoders:*** *Our 5-qubit decoder needs 188 CX gates because it uses no ancilla qubits, so it only pays off when gates are almost ideal.*
- ***No Syndrome Extraction Yet:*** *We decode with a coherent majority gate and do not yet use ancilla qubits or mid-circuit measurement.*

Future Scope

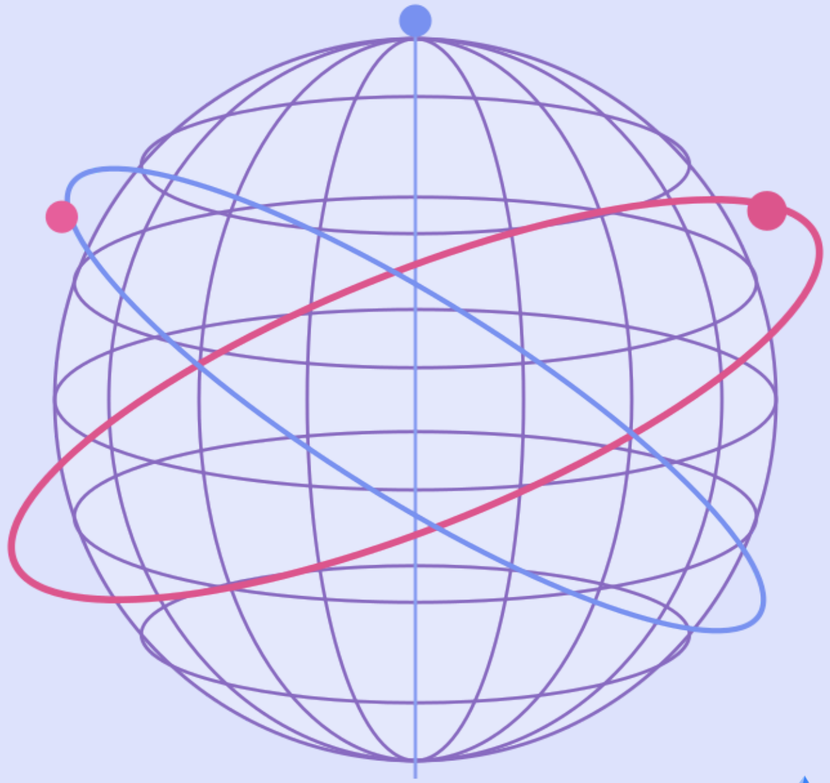
- ***Advanced QEC Codes:*** *Add the Steane, Shor and surface codes for fault-tolerant architectures.*
- ***Syndrome-Based Correction:*** *Add ancilla-based syndrome extraction with mid-circuit measurement and feed-forward.*
- ***Live Hardware Integration:*** *Fetch real noise parameters from IBM Quantum backends through Qiskit Runtime, clearly labelled as hardware-derived, and validate Q-Adapt's recommendations on a real QPU.*
- ***Richer Noise Models:*** *Add correlated errors, amplitude damping and thermal relaxation.*
- ***Smarter Decision Engine:*** *Explore sequential or learning-based selection once more strategies are available.*
- ***Extended Noise Maps:*** *Extend our current p_x–p_z regime map to include gate and read-out noise together on one view.*

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-team-deepspark

DEMO VIDEO LINK
https://youtu.be/3IFeJeFZtSE?si=ZRRzUe2F3Hrg_GKr